ST.ALBERT'SCOLLEGE (AUTONOMOUS),ERNAKULAM

A STUDY ON THE BEHAVIOUR OF GUPPY FISH (Poecilia reticulata)

SUBMITTED BY

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CERTIFICATE

This is to certify that the Behaviour study Report submitted by ALEENA MARIYA Reg No ……………………….is a bonafide record of the student at St. Albert's College (Autonomous), Ernakulam.

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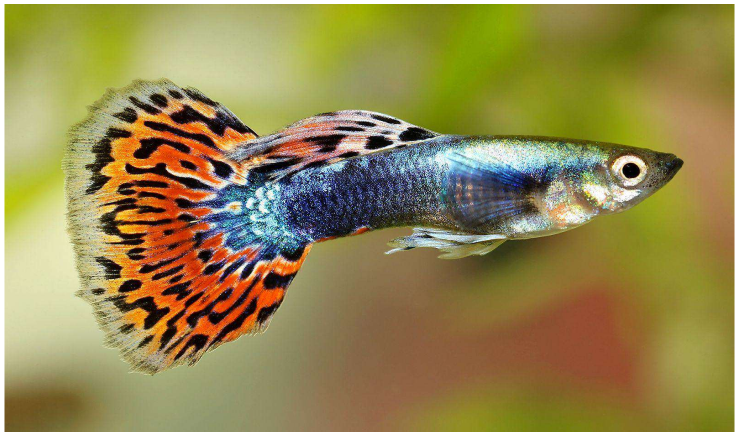
INTRODUCTION

Ethology is the scientific study of animal behavior, typically with a focus on behavior under natural conditions. It seeks to understand how animals interact with their environment, other animals, and members of their own species. Ethologists observe and analyze instinctive behaviors—those that are genetically hardwired—as well as how these behaviors evolve over time. This branch of biology integrates principles from evolution, ecology, and neurobiology to explore why animals behave the way they do and how such behaviors contribute to survival and reproduction.

The field of ethology has been shaped by the work of several pioneering scientists. Konrad Lorenz, often regarded as one of the founders of ethology, introduced key concepts such as imprinting in birds. Nikolaas Tinbergen contributed significantly with his four questions explaining animal behavior from evolutionary, developmental, mechanistic, and functional perspectives. Karl von Frisch, another notable figure, decoded the complex "waggle dance" of honeybees, showing how they communicate information about food sources. Together, these scientists laid the foundation for modern ethology, earning a shared Nobel Prize in 1973 for their groundbreaking work.

Poecilia reticulata, commonly known as the guppy, millionfish, or rainbow fish, is a small freshwater species that belongs to the family Poeciliidae. It is native to the warm freshwater habitats of northeastern South America, but due to its popularity and hardiness, it has been introduced to almost every continent. Guppies are well-known for their striking colors and patterns, especially in males, which are smaller, slender, and brightly ornamented with vivid tails, while females are larger and generally less colorful. One of their most remarkable features is that they are livebearers, giving birth to fully formed, free-swimming young instead of laying eggs. Because of their adaptability, guppies thrive in a wide range of water conditions and are commonly found in streams, rivers, and ponds. They are omnivorous, feeding on algae, plant matter, and tiny invertebrates, which also makes them valuable in controlling mosquito larvae in many parts of the world. Beyond being one of the most popular aquarium fish for hobbyists, guppies are also important in scientific studies, serving as model organisms in research related to evolution, ecology, genetics, and behavior.

GUPPY FISH



Scientific Name: *Poecilia reticulata*

Kingdom: Animalia

Phylum: Chordata

Class: Actinopterygii (ray-finned fishes)

Order: Cyprinodontiformes

Family: Poeciliidae

Genus: Poecilia

Species: reticulata

GENERAL CHARACTERISTICS

Poecilia reticulata, commonly known as the guppy, is a small yet highly popular freshwater fish recognized for its vibrant appearance, adaptability, and scientific importance. In terms of size and appearance, males are generally smaller, measuring about 1.5–3.5 cm, while females grow larger, ranging from 3–6 cm. Males are more strikingly colored, displaying vivid patterns and longer, flowing tails, which they often use in courtship displays, whereas females appear larger, rounder, and comparatively less colorful. These differences between males and females are examples of sexual dimorphism, a key feature in guppies.

Their natural habitat includes freshwater environments such as streams, rivers, and ponds, but they are also well adapted to artificial settings, making them one of the most common fish in aquariums worldwide. Guppies thrive in a variety of water conditions and are highly social creatures that prefer living in groups, often forming shoals for protection and interaction. Their behavior is fascinating to observe, as they not only exhibit shoaling and mating rituals but also demonstrate learning abilities, social recognition, and predator avoidance strategies.

In terms of breeding, guppies are livebearers, which means that females give birth to fully formed, free-swimming young instead of laying eggs. A single female can produce offspring approximately every 30 days, and each brood can contain dozens of fry, making them prolific breeders. This rapid reproductive cycle contributes to their success in both wild and captive environments.

Their diet is omnivorous, consisting of algae, plant matter, insect larvae, brine shrimp, and commercial flake food in aquariums. This flexible feeding habit allows them to adapt to different ecological niches while also serving an important role in controlling mosquito populations by consuming larvae in stagnant waters.

The popularity of guppies extends beyond the aquarium trade. While they are prized as ornamental fish for their beauty and ease of care, they are also extensively used in scientific research. Guppies serve as model organisms in studies of genetics, ecology, evolution, and animal behavior, particularly in understanding mate choice, natural selection, and predator-prey interactions. Because of their adaptability, ease of breeding, and observable traits, they remain one of the most studied freshwater fish species in the world.

Female

Male

EXPERIMENTS

The following experiments were conducted in a familiar home setting to observe and study social behavior, feeding patterns, and learning ability in guppies.These observations provide insights into the ecological and adaptive significance of its behaviour.

EXPERIMENT 1: SHOALING BEHAVIOR

Setup: 5 guppies placed in a tank.

Observation: Guppies prefer to stay in groups, showing strong shoaling tendencies especially when alone or in a new environment.

EXPERIMENT 2: FEEDING BEHAVIOR

Setup: Food was introduced at a fixed corner of the tank.

Observation: Food was introduced consistently at a fixed corner of the aquarium at the same time each day.Over time by repeated trials , guppies learned to associate that corner with food and started gathering there even before feeding time.

EXPERIMENT 3: RESPONSE TO NOVEL OBJECTS

Setup: A Plant was introduced to the tank.

Observation: Guppies initially avoided it but later approached and interacted, suggesting exploratory behavior and learning.

EXPERIMENT 4: GENDER INTERACTION

Setup: Males and females were observed together.

Observation: Males displayed vibrant tail-fanning and chasing behavior to attract females. Females responded base

DISCUSSION

Guppies (Poecilia reticulata) are well-known for their highly social and adaptable behavioral patterns, which have made them a central species in behavioral ecology and ethological studies. One of their most important social behaviors is shoaling, where individuals aggregate in groups. This behavior reduces individual predation risk by creating a "dilution effect," enhances vigilance by allowing collective detection of predators, and minimizes stress through social comfort. In addition, shoaling can improve foraging efficiency, as guppies are more successful in locating and exploiting food sources when in groups compared to when isolated. Such tendencies demonstrate how social structures in guppies contribute to both survival and overall fitness.

Beyond sociality, guppies exhibit notable learning and cognitive flexibility. Research has shown that they are capable of associative learning, as they can remember repeated food placement and adjust their foraging strategies accordingly. Their exploratory responses to novel objects further indicate curiosity and problem-solving capacity, traits that highlight their adaptability to changing environments. These abilities suggest that guppies possess a form of environmental awareness that enables them to optimize survival strategies even under unpredictable conditions.

Reproductive behavior in guppies is equally complex. Males often display vibrant coloration, perform elaborate courtship dances, and use sneaky mating strategies to maximize reproductive success. Females, however, demonstrate selective mate choice, often preferring males with brighter coloration and more vigorous displays, traits that may signal genetic quality and overall fitness. This selective mating preference has long been a subject of evolutionary studies, as it provides insights into sexual selection, reproductive strategies, and mate-choice evolution.

Risk assessment is another significant aspect of guppy behavior. Guppies exhibit plasticity in balancing trade-offs between feeding opportunities and predator avoidance. For example, in high-risk environments, they are more cautious and shoal more tightly, whereas in safer conditions, they may forage more independently. This behavioral flexibility illustrates their ability to modify strategies according to ecological pressures, reinforcing their reputation as a model species for studying adaptive decision-making.

Overall, guppy behavior exemplifies the integration of social cooperation, cognitive ability, reproductive strategy, and ecological adaptability. Their shoaling tendencies, learning capabilities, mate-choice dynamics, and risk assessment align closely with established ethological studies, making them a prime subject for understanding broader concepts in animal behavior, cognition, and evolution.

CONCLUSION

Guppies (Poecilia reticulata) stand out as a species that embodies a unique combination of intelligence, sociality, and behavioral adaptability. Their cognitive abilities, though often underestimated in small freshwater fish, are well-documented through observations of associative learning, spatial memory, and adaptive responses to environmental stimuli. For example, guppies can form associations between feeding locations and reliable food sources, adjust their behavior in response to repeated human interaction, and recognize familiar individuals within a group. Such findings illustrate not only their short-term learning capacity but also their ability to modify long-term strategies in ways that improve survival.

Equally significant are their social dynamics, which demonstrate the evolutionary advantages of group living. Shoaling, in particular, offers multiple benefits: reducing predation risk through collective vigilance, enhancing foraging efficiency by sharing information, and lowering stress through social companionship. These behaviors highlight the guppy’s reliance on cooperation, while also revealing the role of group decision-making and social learning in their daily lives. In addition, the variation in shoal structure and size under different ecological pressures underscores the guppy’s remarkable flexibility in balancing individual and group needs.

Reproductive behavior adds another dimension of complexity. Males exhibit diverse strategies ranging from elaborate courtship displays to opportunistic mating attempts, while females exercise selective mate choice, often preferring brightly colored and energetically displaying males. This interplay of strategies reflects strong sexual selection pressures, driving the evolution of coloration, display behaviors, and even cognitive traits. Moreover, mate-choice copying—a behavior in which females are influenced by the preferences of others—provides further evidence of social learning and the integration of individual and group decision-making.

Risk assessment is another domain where guppies demonstrate advanced behavioral plasticity. In high-risk environments, they tighten shoal formations and prioritize vigilance, whereas in low-risk settings, they allocate more effort to exploration and foraging. This ability to evaluate trade-offs between feeding efficiency and predator avoidance reveals a sophisticated capacity for adaptive decision-making. Such strategies highlight guppies as a species finely tuned to ecological variability.

Taken together, these behavioral traits make guppies an invaluable model organism in ethology and behavioral ecology. Their small size, short generation time, and ease of observation in both laboratory and natural settings allow for a wide range of controlled experimental studies. Researchers have used guppies to explore fundamental questions about cognition, social learning, mate choice, predator-prey dynamics, and evolutionary adaptation. Furthermore, their ability to thrive in both native and introduced habitats offers insights into resilience, plasticity, and the broader mechanisms of ecological success.

Ultimately, guppies bridge the gap between simple vertebrate models and more cognitively complex animals. Their intelligence, social organization, and adaptability not only enrich our understanding of fish behavior but also contribute to the larger framework of animal cognition, cooperation, and evolutionary biology. As a result, guppies continue to provide profound insights into how organisms learn, adapt, and thrive in the face of ecological and evolutionary challenges.

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